

Necessary Existence Without Modal Axioms

What the cosmological arguments buy with a modal logic and a principle of sufficient reason, and what the naturalist buys with a brute fact: reached by elimination from one observation, that non-being requires being

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Abstract

The modern arguments for a necessary being pay in advance. Pruss and Rasmussen pay with the modal system S5 and a causal principle. Leibniz and his heirs pay with a principle of sufficient reason. Leftow pays with a theory of modality grounded in God. Their critics pay too: Oppy with a brute or necessary natural state, the metaphysical nihilist with a modal theory on which an empty world is possible, Maitzen with the dissolution of the question.

An eliminative chain of 137 resolutions reaches what the debate calls a necessary being from one observation, that non-being requires being (R.1), with no modal axiom, no principle of sufficient reason, no explanatory principle, and no theory of modality. What “nothing” designates cannot be (R.2), so being has no alternative. Possibility cannot be limited (R.4). Being comes into being by neither necessity nor randomness, and the survivor, choice, loses its “not” and is Will (R.10, R.11). Will is neither externally justified nor without basis (R.12): the two horns the debate has argued between, sufficient reason and brute fact, both empty. The first actualisation from unlimited possibility is necessarily full (R.42), which answers the naturalist’s necessary initial state and says what the modest arguments leave unsaid.

The chain has no notion of necessity. What the debate calls necessary existence is, on the chain, what remains when every alternative requires what cannot be.

1. What the arguments buy, and what they pay

Leibniz asked why there is something rather than “nothing” and answered with a principle: no fact holds without a sufficient reason. Leibnizian cosmological arguments have used it ever since. It has carried its objectors too, since a reason for everything either includes a reason for itself or does not. Pruss defended the

principle at book length in 2006. Ross, in Goldschmidt's volume, grants that the objections to it fail and argues that the grounds on which they fail take the persuasive force out of the cosmological arguments built on it. Kleinschmidt, in the same volume, gives up the principle for a weaker one about explanatory power.

Pruss and Rasmussen's *Necessary Existence*, in 2018, pays with something else. Six arguments reach a concrete necessary being, concrete meaning capable of causation. Five of them are new, and those five start from possibility premises rather than from a demand for explanation. One of them, the modal argument from beginnings, needs a causal principle only in modal form: for any positive state of affairs that can begin to obtain, it is possible that something external causes it to obtain. With the further premise that contingent concrete things could have begun, and with the system S5, in which what is possibly necessary is necessary, a necessary concrete being follows. The authors are careful about what they have bought. The being is concrete and necessary. The six arguments stop there. The book does not argue that the being is God.

The price is the modal logic and the possibility premises, and the critics have priced them. Oppy's reply, in his chapter in the Goldschmidt volume and across his work, is that the naturalist can pay less for the same goods. If a necessary concrete being is wanted, let the initial state of natural reality be that being. A necessary natural first state explains what a necessary God explains, and the possibility premises that favour the theist over the naturalist are no more evident than the conclusion they serve. O'Connor answers that an ultimate explanation of contingent things needs a being that is not one of them, necessary and transcendent. The argument between them is over which necessary thing to posit. Neither derives that any being is necessary. Both begin by supposing it.

The rest of Goldschmidt's volume shows the other ways to pay. Rodriguez-Pereyra defends the subtraction argument: take away concrete things one by one and a world with none of them is possible, so no concrete thing is necessary. Efrid and Stoneham observe that the verdict turns on which theory of modality one holds. Lowe argues the other way, that the existence of abstract objects entails that contingent objects must exist. Kotzen examines the probabilistic answer van Inwagen gave, that an empty world is one possibility among infinitely many and so has no chance, and finds that the required probability assignments cannot be justified. Maitzen argues that the question itself is confused, since ordinary things already explain why there are things, and no ultimate explainer is wanted. Leslie proposes that the universe exists because it is good that it exists, and that goodness is creatively efficacious. Leftow, in *God and Necessity*, proposes that modal truth itself is grounded in God, so that God is the one necessary being and everything else is necessary or possible because of him. Kuhn's taxonomy of nine "nothings," from empty space to an absolute "nothing" with no laws and no possibilities, shows how many different things the word has been made to mean.

Each of these positions wants the same thing: a necessary being, or a reason why none is needed, without begging the question. Each pays for it with something the opponent can refuse: an axiom of modal logic, a principle of sufficient reason, a possibility premise, a probability assignment, a modal theory, an axiological principle, a brute fact.

The chain pays with none of them. It begins with one observation the parties to this debate share, that non-being requires being, and reaches by elimination that “nothing” cannot be, that possibility cannot be limited, that being comes into being as Will, that Will wills as itself, neither externally justified nor without basis, and that the first actualisation is necessarily full. Each step names a candidate, tests it against what earlier steps established, and keeps only what survives. The chain has no notion of necessity. Where R.10 names Necessity, an outcome forced by a constraint, it is a candidate for how being comes into being, and it dissolves. What the debate calls necessity is, on the chain, the standing of what remains when every alternative requires what cannot be.

There is a derivation. It is set out in a sequence of 137 resolutions titled *Anything Willing Is*, and its first fourteen steps are presented in academic prose in a short paper, *The Eliminative Derivation of Will*. This paper does not repeat either. It says where each position in the debate sits on the chain, what the chain reaches that the position wanted, and what it refuses that the position paid. Every number below is an anchor on the interactive edition, so any step can be read with its own test. At five places the test itself is also set out in a sentence.

2. “Nothing” cannot be

The chain’s first three resolutions settle the question Leibniz asked, and settle it by removing its contrast class. Non-being is the “not” of being. Without being there is no negation, so non-being requires being (R.1). The word “nothing” names non-being in the absence of being: the “not” of what is not there to be negated. Non-being requires being, so non-being without being is a “not” with no being to be the “not” of, and what the label names cannot be (R.2). Without the possibility of being, being cannot be, because what remains without that possibility is the absence of being, which is what “nothing” designates (R.3).

This is the first place the test belongs on the page, because it is the chain’s foundation and everything after it descends from it. The candidate at R.2 is “nothing” as something that could be. “Nothing” is non-being without being. Non-being requires being (R.1). So “nothing” is non-being without what non-being requires. The candidate requires what its own content forbids, and it dissolves. There is no premise here. R.1 is itself an elimination: its candidate, non-being without being, dissolves because negation requires what it negates.

Leibniz’s question, why something rather than “nothing,” presupposes that the word names an alternative. R.2 says it does not. The question does not go unanswered. It loses its second term. This is close to Maitzen’s conclusion that the question is confused, but it reaches the conclusion differently and keeps more.

Maitzen argues from the adequacy of ordinary explanations. The chain argues from what “nothing” means, and it does not stop at dissolving the question: it goes on to say what has no alternative and why, which Maitzen’s deflation cannot.

Kuhn’s nine “nothings” sort themselves at R.2. Every one but the last leaves something in place: empty space, or fields, or laws, or abstract objects, or possibilities. On the chain each of those is non-being with being, a “not” of some things with other things remaining to be the “not” of. Only Kuhn’s absolute “nothing,” with no possibilities, is what R.2 names, and R.2 says that one cannot be. The taxonomy is right that the word has been made to do nine jobs. The chain says which job is impossible and why the other eight are not “nothing.”

The subtraction argument sorts itself the same way. Rodriguez-Pereyra’s conclusion is that a world with no concrete objects is possible. The chain does not need to dispute the subtraction. A world with no concrete things still has the possibility of being. R.3 says being cannot be without it, and the possibility is not a concrete thing. So the empty world of the subtraction argument is not “nothing” in R.2’s sense. Whether such a world could have been is not a question the chain asks. Being is (R.1). The chain asks how being comes into being from the unlimited (R.10) and what the first actualisation must be (R.42). Its first result about the subtraction argument’s world is only that it is not “nothing.” Lowe’s argument that abstract objects require concrete ones reaches the anti-nihilist conclusion the chain also reaches, from a different dependency. Kotzen is right that the probabilistic answer fails, and the chain shows why it was never needed. That answer treats the absence of all being as one possibility among infinitely many. On the chain the absence of all being is not a possibility of any probability: it is what “nothing” designates, and that cannot be (R.2). Efid and Stoneham are right that the subtraction verdict depends on one’s theory of modality. The chain has no theory of modality, which is the subject of section 6.

3. Possibility cannot be limited

The chain’s next result is the one the modal arguments assume and never derive. If possibility is limited, then beyond the limit there is no possibility of being. Without the possibility of being, being cannot be (R.3). What is beyond the limit is therefore what “nothing” designates, which cannot be (R.2). The limit requires, beyond itself, what cannot be. Possibility is without limitation, and this is what the word “anything” points to (R.4, Def.1).

This is the second place the test belongs on the page. The candidate is limited possibility. A limit on possibility is a boundary beyond which being is not possible. Beyond that boundary, being cannot be (R.3), and the region beyond it is the absence of being, which is what “nothing” designates (R.2). A limit on possibility requires what R.2 establishes cannot be. It dissolves. One conse-

quence is worth stating because it is the chain’s answer to a natural objection: “a limit on possibility is just some things not being possible.” The objector’s “not” is the “not” of what is (R.1). A square is not a circle because each is what it is. That is specificity, not a boundary on possibility, and specificity is not restriction (R.11). A boundary beyond which being cannot be would have what “nothing” designates on its far side, and that cannot be (R.2, R.3).

Three more results follow before being enters. Unlimited potential does not include negation, because negation is exclusion, exclusion is restriction, and restriction empties “unrestricted” (R.7). What is, is a something, and what the word “anything” points to is not a something (R.8). Unlimited potential includes the potential for a something to come into being, since a potential that lacked it would lack something and would not be unrestricted (R.9).

Beside R.4 stands S5. The system’s characteristic axiom is that what is possibly necessary is necessary, and its use in the argument from beginnings is to carry a possibility to an actuality. The chain never uses the axiom, because it never has a bare possibility in hand. Its possibility is possibility of being (R.3), reached from R.2, and its unlimitedness is reached at R.4 by elimination, not assumed as a feature of a modal space. There is no space of possible worlds in which a necessary being either exists in all or none. There is possibility without limitation, and there is what it actualises. Pruss and Rasmussen’s modal uniformity principle, that similar things have similar modal profiles, is a reasonable heuristic within a space of possible worlds, where a thing’s profile is the set of worlds it belongs to. The chain has no worlds. It has one possibility, without limitation (R.4), and what is actualised from it. Where there is no space of worlds, there are no profiles to be uniform. What possibility includes is answered by what it cannot exclude.

4. From possibility to being, without a principle of sufficient reason

Being is (R.1), and being is specific (R.5): to be is to be something. Unlimited potential includes the potential for specifics (R.9). How does a specific something come into being from unlimited possibility? Its specificity either has a source or does not. If it does not: randomness. If it does, the source is either external to what is specific or internal to it. External: necessity. Internal: choice. The three exhaust the options (R.10).

This is the third place the test belongs on the page, and it is the one that replaces the principle of sufficient reason. Necessity: the outcome is forced by constraint, constraint is restriction, and restriction empties “unrestricted” (R.4). Randomness: two routes dissolve it. First, under randomness the absence of any principle would itself have to determine the transition. An absence is the “not” of what is present (R.1). Under randomness no principle is present, so the absence has no being to be the “not” of, and an absence with no being to be the “not” of is what “nothing” designates, which cannot be (R.2). Second, un-

der randomness no principle excludes any outcome, including non-actualisation, and non-actualisation is negation of actualisation; unlimited potential does not include negation (R.7). The defence, randomness with actualisation guaranteed but the specific outcome random, fails on both arms of its own division: the guarantee is a constraint, which is necessity as to whether, and as to which the selection would again be by an absence with no being to be the “not” of. What remains is the something’s own: choice.

Three things have not been used. No principle that every contingent thing has an explanation. No principle that every state that can begin can be caused. No possibility premise. The randomness horn is not refused because it violates a demand for reasons. It is refused because the absence it asserts is, by R.1 and R.2, a “not” with no being to be the “not” of. That is why Oppy’s brute fact is already met here, before Will is named. The naturalist who says that the initial state simply is, with no reason, asserts an absence of principle. The assertion is the same absence R.10 tests, and it dissolves the same way. It is not that brute facts are forbidden by a rationalist axiom. It is that a brute fact, stated, names what cannot be.

Choice survives R.10 in the form the word ordinarily has: “this, not that.” At R.11 the “not that” does not survive. “Not that” requires something to be excluded, and in unlimited potential exclusion empties “unlimited” (R.4). The defence that the exclusion is local, leaving “that” fully available, conflates two things: not actualising “that,” which leaves potential open and is compatible with the unrestricted, and actively walling “that” off, which restricts and is not. What remains of choice is “this is,” without “not that,” and the chain names it Will: actualisation without exclusion (Def.2). Each “this is” adds to what is without removing from what could be.

Then R.12, and this is the fourth place the test belongs on the page, because it is where the two sides of the debate have stood since Leibniz. Will wills as itself: not externally justified, and not without basis. If Will requires an external justification, that justification constrains Will, an external constraint is restriction, and restriction empties “unrestricted” (R.4). If Will wills without any basis, actual being is specific (R.5) but what makes it what it is, is not, and that empties “specific.” Neither survives. The willing is the willing (R.12, Def.3).

The principle of sufficient reason is the first horn. It asks that the first actualisation be justified by a reason outside itself, and a reason outside Will constrains it. The brute fact is the second horn. It asks that the first actualisation have no basis, and a specific being with no basis for being what it is empties the specificity it must have. The debate between Pruss and Oppy has mostly been conducted as if these were the only two positions: either everything has a sufficient reason or something is a brute fact. R.12 is the position between them, and it is reached by dissolving both. Will wills as itself. The willing is the willing. That is not a compromise. It is what remains. Ross is right that the objections to the principle of sufficient reason fail, and right that this does not

rescue the arguments built on it, because the arguments needed the principle to do work the chain does with R.4 and R.5 alone. Kleinschmidt's explanatory-power principle is not needed either. No principle about explanation appears on the chain. What appears is a partition of the ways a something could be specific, and two of the three arms require what cannot be.

5. What the necessary being is

Pruss and Rasmussen stop at a concrete necessary being and do not say what it is. Oppy fills the role with a necessary natural state. Leftow makes it God by making modality depend on it. The chain continues.

Will establishes order, because the first willing is the first differentiation and there is no order before differentiation (R.13). Will does not negate; negation requires what already is and enters only when what is wills against what is (R.14). Differentiation and unity are inseparable in what the word "anything" points to, since the Will that makes being distinct is the Will being is (R.15). The being is its willing (R.16). All being is Will (R.17). Actualisation preserves the unrestricted, because unrestricted potential is not a store depleted by use (R.18).

Then R.42, and this is the fifth place the test belongs on the page, because it is where Oppy's necessary natural state is met. The first actualisation from unlimited potential is necessarily full. Partiality is "this, not that," being this and not what is excluded. R.11 eliminated "not that" from unlimited potential. Without "not that," the first actualisation cannot be partial: it is everything unlimited potential includes, because exclusion, the only means by which an actualisation could be less than everything, does not survive R.11. Two defences are tested. That the first actualisation limits itself: self-limitation is exclusion, and exclusion did not survive. That it is simply a specific nature, less than everything without any exclusion: a specific nature that is not everything requires a limit that is not exclusion, and the only candidate is that the nature simply stops, which is ceasing to generate, which empties "Will is generative" (R.24). Both dissolve. The first actualisation is full. It is All-Being (R.39, R.42), Anything's nature fully operative as being (R.40). Its willing is every being's willing, one Will and not a set of willings (R.44), and the unity is not coercive (R.19).

A necessary natural first state is a partial necessary being. It is this configuration and not others. On the chain that partiality is exclusion, and exclusion has no standing in unlimited potential. So the naturalist's candidate does not fail because it is natural. It fails because it is less than everything, and the first actualisation from the unlimited cannot be less than everything. Oppy is right against Pruss and Rasmussen that a naturalist who takes the initial natural state to be necessary meets their conclusion as well as a theist does, so their arguments do not say why the necessary being should be God rather than a first natural state. The chain says why it is neither, as those terms are used in

the debate: it is the full actualisation, and what it is called comes later. The manuscript's convergence table sets All-Being beside the name theology gives it, after the derivation and not before.

This also says what the modest conclusion left unsaid. Pruss and Rasmussen's being is concrete, meaning capable of causation. The chain's is Will, and Will is generative (R.24), which is more than capable of causation and is reached rather than defined. Their being is necessary in that it cannot fail to exist. The chain does not say necessary. It says that being has no alternative (R.2) and that the first actualisation cannot be less than everything (R.42). And where their being might be one of many possible necessary beings, the chain has one: another full actualisation would be indistinguishable from the one there is, and is it (R.43). No omnipotence, omniscience, or goodness is assumed. What the chain reaches at R.39 and R.40 is willing present to all reality with all reality known, and what it reaches at R.47 is that generating is flourishing. Leslie's axiarchism, that the universe exists because it is good that it exists, has its counterpart here with the order reversed. Goodness is not a principle that brings being about. It is what the chain finds Will to be: generating is flourishing (R.47), and the convergence table names Anything's nature Good. Goff's cosmic purpose, reached from fine-tuning and consciousness, has the same counterpart: purpose is not added to a universe that lacked it. Generating is increase, and knowing is being (R.28). The chain arrives where Leslie and Goff want to arrive, from the other end, and does not need their premises.

6. Necessity without a theory of modality

Leftow's *God and Necessity* sets out to say what necessity is, and it says that necessary truths are grounded in God: some in God's nature, the rest in what God freely brings about. The proposal answers a real question that the possibility-based arguments leave open. If S5 is a truth about a space of worlds, what makes it true, and what makes the worlds? Leftow's answer is God. The cost is a theory of modality that the arguments for God then depend on.

The chain has no theory of modality, and this is not an omission. What the debate would call its necessity is a standing, not an operator. A result stands when every alternative requires what an earlier result established cannot be, and at the base that is R.2: what "nothing" designates cannot be. "Cannot" on the chain never means "false in all possible worlds." It means "requires what R.2 establishes cannot be," or "empties an established result of its content." The eliminations at R.4, R.10, R.11, R.12, and R.42 are of this kind. No result rests on a ground. What holds, holds because its denial names the impossible.

This is why the chain can grant Efrd and Stoneham their point and be untouched by it. They are right that whether an empty world is possible depends on one's modal theory. The chain does not have one and does not decide the question by one. It decides it at R.2 and R.42: "nothing" cannot be, and the first actualisation is full. And it is why the chain can meet Leftow's question

without his answer. What makes possibility unlimited is not a divine nature or a divine act. It is that a limit would require what cannot be (R.4). What makes the first actualisation necessary is not that it exists in every world. It is that being has no alternative (R.2) and that the first actualisation, being given (R.1), cannot be partial (R.11, R.42). Leftow makes God the ground of modality. The chain reaches All-Being and has no modality to ground.

Two objections to the method belong here. The first is that R.1 is a premise, and the chain therefore has an axiom after all. It is not a premise. It is the first elimination. Its candidate is non-being without being, and the test is the content of the word “not”: non-being is the negation of being, and without being there is no negation. Whoever denies R.1 asserts a “not” with no being for it to be the “not” of, which is what R.2 finds cannot be. The denial cancels itself. That, and not agreement, is the base’s standing, and it is why no premise is added anywhere after it. The second objection is that the method judges itself: eliminations are themselves claims, and a claim that “nothing” cannot be is as contestable as any other. It is not. Its denial, that “nothing” can be, is the candidate R.2 tests, and it dissolves on the content of its own words. A method whose first result stands by the self-cancellation of its denial can be applied to its own steps. Each is a candidate, and each stands or falls by the same test.

What has been reached is two necessities from one observation. The first is that being has no alternative, because what “nothing” designates cannot be. The second is that the first actualisation from unlimited possibility is everything that possibility includes, because it cannot be less. Neither uses a modal logic, a principle of sufficient reason, a possibility premise, or a theory of modality, and neither posits a brute fact. The debate has paid for a necessary being in one of those currencies since Leibniz. The chain reaches it, and says what it is, without paying.

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